ANALYTICAL CUMULATIVE EXAM

October 5, 1991

One of the most exciting developments in the past seven years has been the demonstration that electrophoretic separations could be performed in narrow capillary tubes (10-100 μm internal diameter) 80-100 cm in length with applied voltages ranging from 30 to 40 kV/m (Science, **1983**, 222, 266). This technique is known as capillary electrophoresis (CE). Exceptionally high separation efficiencies have been demonstrated for both charged and uncharged molecules.

a. (20) In any electrophoretic separation, negatively charged species will migrate toward the anode, while positively charged species migrate toward the cathode driven by their electrophoretic mobility. In CE however, all ions, independent of charge migrate toward one electrode making automated detection easier. Describe in detail the origin of the forces that lead to this phenomenon, that is, the net movement of both cations, neutrals and anions towards the cathode.

b. (15) Sample volumes required for CE are in the 1 to 10 nL range. Describe one technique for introducing such small volumes onto a CE column. Briefly describe any limitations of the technique.

c. (15) Sample elution zones in CE are less than 100 nL. Describe one detector system compatible with these small elution volumes.

d. (35) Giddings has suggested that one measure of efficiency in CE systems is the number of theoretical plates, N, where

$$N = L^2/\sigma^2$$

Here, L is the capillary length and σ^2 the peak variance. The variance is described by the Einstein equation when molecular diffusion alone is responsible for zone broadening.

$$\sigma^2 = 2Dt$$

Here D is the diffusion constant and t is time. For migration of a charged species with an electrophoretic mobility of μ in response to a field of V total volts, the velocity of migration is given by

$$v = \mu V/L$$

Using the conventional definition of velocity in terms of distance traveled per time, derive an expression for N in terms of the electrophopretic mobility, the applied voltage and the diffusion constant.

e. (15) From your expression, calculate the gain in efficiency in going from an applied voltage of 1000 V to 40 kV.